

HAFIZH FADHL MUHAMMAD

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PROFILE SUMMARY

A 6th-semester Computer Science student at Universitas Padjadjaran with a GPA of 3.78/4.00 and a strong interest in Data Science, Machine Learning, and Natural Language Processing (NLP). Experienced in academic and independent projects involving predictive modeling, sentiment analysis using IndoBERT, and time-series forecasting with XGBoost. Skilled in Python, SQL, Scikit-learn, and PyTorch, with hands-on experience in data preprocessing, feature engineering, and model evaluation. Known for being analytical, detail-oriented, and research-oriented, with strong problem-solving abilities and a passion for developing data-driven solutions. Eager to apply technical and analytical skills in real-world environments through internship opportunities in Data Science.

EDUCATION

Padjadjaran University

Bachelor of Computer Science

Aug 2023 – Present

- **Current GPA:** 3.78/4.00 (109 credits)
- **Relevant Coursework:** Data Mining, Artificial Intelligence, Statistics, Database Systems, Decision Support Systems, Numerical Methods, Linear Algebra.
- **Achievements:**
 - o Selected as Team Leader of Universitas Padjadjaran Delegation (“Warlok” Team) for GEMASTIK 2025 Data Mining Competition.
 - o Semifinalist – Business Plan Competition by IFEST (Oct 2025).

Bootcamp Data Science & AI Machine Learning

Dibimbing.id

Mar 2026 – Present

- Learning end-to-end data science workflows including data collection, exploratory data analysis (EDA), supervised and unsupervised learning, and model evaluation using real-world datasets.
- Developing practical skills in Natural Language Processing (NLP), time-series forecasting, deep learning, and model deployment for data-driven problem solving.
- Gaining hands-on experience with Python (Scikit-learn, TensorFlow/PyTorch), SQL, and big data tools for analyzing and processing large-scale datasets.

WORK EXPERIENCE

Self-Employed

Remote

Freelance Data Scientist

Jan 2026 – Present

- Engineered 10+ end-to-end data science solutions by performing rigorous data preprocessing and exploratory data analysis (EDA) on datasets exceeding 100,000+ records.
- Developed and Evaluated machine learning models for complex classification and prediction tasks using Python and Scikit-learn, consistently improving performance through advanced feature engineering and hyperparameter optimization.
- Architected structured data queries using Python and SQL to extract actionable insights, directly supporting clients in making strategic, data-driven decisions.

Pusat Inovasi Pengajaran dan Pembelajaran (PIPP), Universitas Padjadjaran

Jatinangor, Indonesia

Data Scientist Intern

Sept 2025 – Dec 2025

- Developed machine learning and deep learning pipelines for EEG-based emotion classification by processing 1,000+ signal segments and applying feature extraction techniques.
- Built and Evaluated 5+ classification models, including SVM, Random Forest, XGBoost, and MLP, using LOSO and LOVO cross-validation to improve model robustness and generalization performance.
- Improved model performance by 15–30% through data preprocessing, feature engineering, and comparative model evaluation techniques.

Universitas Padjadjaran

Jatinangor, Indonesia

Laboratory Teaching Assistant

Feb 2025 – Dec 2025

- Mentored 40+ students per semester in Object-Oriented Programming (Java), Database Systems, and Web Programming through structured laboratory sessions and technical guidance.
- Guided students in SQL database design, debugging, and problem-solving to strengthen programming and database management skills.
- Evaluated 100+ technical assignments and provided constructive feedback to support student learning and technical improvement.

ID/X Partners

Remote

Data Scientist (Project-Based)

Jan 2025 – Feb 2025

- Analyzed 466K+ financial loan records through rigorous data cleaning and exploratory data analysis (EDA) to identify borrower risk patterns and support credit risk assessment.
- Developed and Benchmarked 6 machine learning models, including XGBoost and Random Forest, to predict loan default risk, optimizing for predictive performance and model robustness.
- Synthesized actionable insights from feature importance evaluation to support data-driven decision making and strategic risk management in financial lending scenarios.

ORGANIZATIONAL EXPERIENCE

HIMATIF FMIPA UNPAD

Head of Interest and Talent

Jan 2025 – Dec 2025

- Led member development initiatives and supervised 8 staff members across 3 departmental programs to strengthen organizational engagement and talent development.
- Initiated a free-access sports development program (KKM's Care) that increased participant attendance by 100% (from 10 to 20 members) by improving accessibility.
- Developed and Managed the department's Instagram platform from scratch, publishing 67 content posts to expand external visibility and promote student talent initiatives.
- Supervised the execution of a flagship tournament (INFORMATICSGRAM), coordinating 71 committee members and achieving a 12% increase in participants (250 to 280) through large-scale publication strategies.

BEM FMIPA UNPAD

Staff of Data Research and Analysis Bureau

May 2025 – Feb 2026

- Analyzed survey data from multiple organizational programs involving 50+ respondents per survey using Excel and Python to evaluate effectiveness and satisfaction.
- Performed comprehensive data cleaning and preprocessing for nearly 10 internal surveys, identifying key behavioral insights and opportunities for program optimization.
- Generated analytical findings and evaluation reports that served as the primary basis for data-driven decision making and internal program performance improvements.

PROJECTS

- **EEG Emotion Classification (Deep Learning vs Machine Learning):** Engineered an end-to-end classification pipeline using PSD feature extraction and benchmarked 4+ ML/DL models with LOSO/LOVO validation, achieving a 51.26% subject-independent accuracy baseline.
- **Public Sentiment Analysis on "Whoosh" Corruption Issue:** Developed an Indonesian NLP pipeline to analyze 900+ scraped YouTube comments, comparing fine-tuned IndoBERT and TF-IDF models to achieve a 65.66% accuracy and a 0.4527 Macro-F1 score for imbalanced sentiment classification.
- **Food Price Forecasting using XGBoost for Multivariate Time-Series Analysis:** Engineered a multivariate forecasting pipeline for 443K+ food price records across 34 provinces, optimizing performance with Optuna and lag-based feature engineering to achieve a highly precise MAPE of 0.054.
- **Neuro-Fuzzy Credit Risk Prediction with Genetic Algorithm Optimization:** Developed a three-stage credit risk system on 32K+ financial records, optimizing a Mamdani Fuzzy Inference System (FIS) using Genetic Algorithms to improve accuracy from 43.73% to 67.90%, while benchmarking against an ANN that achieved 84.06%.
- **Student Well-Being Clustering using Unsupervised Learning:** Developed an unsupervised learning pipeline to segment student well-being profiles using K-Means, Hierarchical Clustering, and DBSCAN, leveraging PCA and UMAP to identify behavioral patterns for data-driven intervention strategies.

SKILLS

Hard Skills:

- **Machine Learning:** supervised & unsupervised learning, classification, clustering, time-series forecasting, feature engineering, model evaluation.
- **Deep Learning & NLP:** neural networks, transformer-based models, sentiment analysis, text preprocessing, IndoBERT fine-tuning.
- **Statistical & Data Analysis:** exploratory data analysis (EDA), predictive modeling, hypothesis testing, data-driven decision making.
- **Data Wrangling & Processing:** data cleaning, preprocessing, transformation, and feature extraction using Python (Pandas, NumPy).
- **Data Visualization:** data storytelling and insight visualization using Matplotlib, Seaborn, and Power BI.

Toolset: Python, SQL, Scikit-learn, PyTorch, TensorFlow, XGBoost, Hugging Face Transformers, Jupyter Notebook, Git, Streamlit, Power BI, Google Colab.

Languages: Indonesian (Native), English (Upper-Intermediate).

TRAINING AND CERTIFICATIONS

- **Building Deep Learning Applications with Keras Certificate** – LinkedIn Learning (Apr 2026)
- **Machine Learning** – Dicoding Indonesia (Apr 2026)
- **Artificial Intelligence** – Dicoding Indonesia (Apr 2026)
- **SQL (Advanced) Certificate** – HackerRank (Feb 2026)
- **Python For Data Science Certificate** – MySkill (Jan 2026)
- **Data Science Fundamental Certificate** – DQLab (Jan 2024)
- **Python Fundamental For Data Science Certificate** – DQLab (Jan 2024)
- **Alibaba Cloud Certificate** – Alibaba Cloud (Mar 2024)
- **Data Science Intermediate Certificate** – Coding Studio (Jan 2023)